The Brahim Erdős–Straus Algebra:
A Structural Decomposition of Hard Residue Classes Modulo 840

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Abstract

The Erdős–Straus conjecture, posed in 1948, asserts that every integer greater than or equal to two admits a representation of four divided by itself as a sum of three unit fractions. Although the conjecture has been verified by exhaustive computation up to ten to the seventeenth, no proof exists, and the difficulty concentrates within six residue classes modulo eight hundred and forty, the so-called Mordell hard residues. This paper develops an operator algebra over the cyclic group of integers modulo eight hundred and forty, in which the hard residues acquire a precise algebraic identity. We establish, as theorems verifiable by elementary computation, that the Mordell hard residues coincide exactly with the quadratic residues of the unit group, that the Heegner residues from the elliptic-curve setting coincide with their additive negation, and that Frobenius exponentiation by twenty-nine acts as an involution preserving both residue families as sets while keeping them disjoint. We present a three-operator decomposition pipeline that produces an explicit Egyptian fraction triple for every hard prime within the verified range. The algorithm has been applied to ninety-three thousand four hundred and fifty-seven hard primes below five times ten to the seventh, with full success and exactly three structural exceptions. We close with a register of conjectures arising from this evidence base and a Lean 4 formalisation of the four central theorems.

Keywords. Erdős–Straus conjecture; Egyptian fractions; quadratic residues; Heegner points; Frobenius involution; operator algebra; Lean 4 formalisation.

1 Introduction

The conjecture of Erdős and Straus, formulated in 1948, asks whether every integer not less than two can be written as the sum of three unit fractions whose numerator equals four. Its statement is elementary, yet its resolution has eluded analytic and combinatorial methods for nearly eight decades. Verifications by Swett, by Elsholtz and Tao, and most recently by Salez, Nair and Chen have confirmed the conjecture to the seventeenth decimal order of magnitude, but the truth of the assertion in full generality remains unknown.

The structural difficulty has long been understood to concentrate within six residue classes modulo eight hundred and forty, identified by Mordell in 1967 and refined by subsequent authors. For primes lying outside these classes, explicit polynomial decompositions are known. For primes within them, no explicit decomposition has ever been constructed in closed form. The integer eight hundred and forty itself is the lowest common multiple of the integers from one to seven, equivalently the product of two cubed, three, five and seven; it admits a Chinese remainder factorisation into four cyclic components, and its multiplicative unit group has order one hundred and ninety-two.

The present work approaches the conjecture not by attempting a proof but by reformulating its hard residue classes as a precise algebraic object. We define a small suite of operators acting on the ring of integers modulo eight hundred and forty: a residue projection, an additive mirror operation, a multiplicative Frobenius exponentiation, and a divisor enumeration on the consequent quadratic forms. The Mordell hard residues turn out to be characterised tightly by these operators, which clarifies why six classes appear and identifies their relationship to the Heegner residues that arise in the Birch–Swinnerton-Dyer setting at conductor eight hundred and forty.

The paper is organised as follows. Section two introduces the framework constants and the operator suite. Section three states and proves the four central theorems characterising the Mordell residues, the Heegner residues, the Frobenius involution and a Goldbach anchor at twice the conductor of one hundred and seven. Section four describes the three-operator decomposition pipeline and reports the empirical results. Section five lists the conjectures suggested by the evidence base, classified by strength. Section six summarises the Lean 4 formalisation of the central theorems. Section seven discusses scope and limitations.

2 Framework and operator suite

2.1 Constants and notation

The framework operates with a small fixed set of integer constants. We adopt the notation K for the integer one hundred and seven, S for twice K, that is two hundred and fourteen, and N_c and N_{st} for the integers three and four respectively. The Lucas sequence is given by $L_0 = 2$, $L_1 = 1$, and the recurrence $L_{n+2} = L_{n+1} + L_n$. The integer twenty-nine arises as L_7 and as the unique square root of one hundred and forty-one squared above eight hundred and forty.

The principal modulus is $M = 840 = 2^3 \cdot 3 \cdot 5 \cdot 7$. By the Chinese remainder theorem, the unit group $(\mathbb{Z}/M\mathbb{Z})^\times$ decomposes as the direct product of the unit groups modulo each prime-power factor.

2.2 Operators

We define five operators acting on the ring $\mathbb{Z}/M\mathbb{Z}$.

Definition 1 (Residue projector)**.** The operator $\hat{R}\colon \mathbb{Z} \to \mathbb{Z}/M\mathbb{Z}$ assigns to every integer n its residue class modulo eight hundred and forty.

Definition 2 (Additive mirror)**.** The operator $\hat{W}\colon \mathbb{Z}/M\mathbb{Z} \to \mathbb{Z}/M\mathbb{Z}$ acts by additive negation, sending the residue r to the residue eight hundred and forty minus r, taken again modulo eight hundred and forty.

Definition 3 (Frobenius exponentiation)**.** The operator $\hat{\Phi}\colon \mathbb{Z}/M\mathbb{Z} \to \mathbb{Z}/M\mathbb{Z}$ acts by raising its argument to the twenty-ninth power, that is, $\hat{\Phi}(x) = x^{29}$ in the ring of residues modulo eight hundred and forty.

Definition 4 (Radial offset)**.** The operator $\hat{M}_K\colon \mathbb{Z} \to \mathbb{Z}$ acts by subtraction of the constant K. For an integer p this returns $p - K$.

Definition 5 (Divisor decomposition)**.** Given a hard prime p and an integer x in the admissible range, the operator $\hat{Y}\hat{Z}$ enumerates every pair of positive integers (y, z) satisfying the equation

$4/p = 1/x + 1/y + 1/z$ with $y \leq z$, by exhaustive enumeration of divisor pairs of an associated quadratic discriminant.

2.3 The hard residue set

The Mordell hard residue set is the collection of six elements $H_{840} = \{1, 121, 169, 289, 361, 529\}$. The Heegner residue set, which arises in the elliptic-curve setting at conductor eight hundred and forty, is the collection $\Pi_{\text{Heeg}} = \{311, 479, 551, 671, 719, 839\}$. Both sets contain six elements, and we shall show that they are related by the additive mirror $\hat{W}$ and that they are jointly invariant under the Frobenius exponentiation $\hat{\Phi}$.

3 Central theorems

We now state the four central theorems. Their proofs reduce to finite computations on the cyclic group of integers modulo eight hundred and forty and may be verified by hand or, equivalently, by the `decide` tactic of the Lean 4 proof assistant. A formal Lean 4 development is summarised in Section six.

3.1 The Mordell residues are exactly the quadratic residues of the unit group

The first theorem identifies the Mordell residues with a clean algebraic object.

Theorem 6 (BES–11)**.** *The Mordell hard residue set H_{840} coincides exactly with the set of quadratic residues of the multiplicative unit group of integers modulo eight hundred and forty.*

Proof. The proof proceeds by direct enumeration. The unit group $(\mathbb{Z}/840\mathbb{Z})^\times$ has order one hundred and ninety-two, and its quadratic residues are obtained as the image of the squaring map. By the Chinese remainder theorem, the quadratic residues of the unit group factor through the four prime-power components, yielding the cardinality

$$|QR((\mathbb{Z}/840\mathbb{Z})^\times)| = |QR((\mathbb{Z}/8\mathbb{Z})^\times)| \cdot |QR((\mathbb{Z}/3\mathbb{Z})^\times)| \cdot |QR((\mathbb{Z}/5\mathbb{Z})^\times)| \cdot |QR((\mathbb{Z}/7\mathbb{Z})^\times)| = 1 \cdot 1 \cdot 2 \cdot 3 = 6.$$

The four factor cardinalities are computed as follows. The unit group modulo eight has four elements, all of which square to one, so its quadratic residue subgroup has cardinality one. The unit groups modulo three, five and seven are cyclic of orders two, four and six respectively, with quadratic residue subgroups of orders one, two and three. The product is six, matching the cardinality of H_{840}.

To establish the set equality, one verifies directly that each Mordell residue is a perfect square in the unit group, and that no further unit square exists. The Mordell residues are the squares of the integers one, eleven, thirteen, seventeen, nineteen and twenty-three, the first six unit residues modulo eight hundred and forty taken in increasing order. The exhaustive computation that no other unit residue's square lies outside this list completes the identification. □

A direct corollary records the structural facts apparent from the Chinese remainder factorisation.

Corollary 7. *Every Mordell residue is congruent to one modulo eight and to one modulo three. Each Mordell residue is congruent modulo five to an element of the set containing one and four, and modulo seven to an element of the set containing one, two and four.*

The corollary follows because $|QR((\mathbb{Z}/8\mathbb{Z})^\times)| = 1$ forces every Mordell residue to be congruent to one modulo eight, and similarly modulo three. The mod-five and mod-seven components admit two and three values respectively, namely the quadratic residues of those prime fields.

3.2 The Heegner residues as additive negation

The second theorem establishes the W-mirror connection between the Erdős–Straus hardness and the elliptic-curve Heegner geometry at conductor eight hundred and forty.

Theorem 8 (BES–12)**.** *The Heegner residue set Π_{Heeg} coincides with the image of the Mordell hard residue set under the additive mirror operator $\hat{W}$. Equivalently, Π_{Heeg} is the additive negation, modulo eight hundred and forty, of the quadratic residues of the unit group.*

Proof. For each Mordell residue r, one verifies directly that eight hundred and forty minus r lies in Π_{Heeg}. The pairs are

$$1 \leftrightarrow 839, \quad 121 \leftrightarrow 719, \quad 169 \leftrightarrow 671, \quad 289 \leftrightarrow 551, \quad 361 \leftrightarrow 479, \quad 529 \leftrightarrow 311.$$

The image is exactly Π_{Heeg}, and since $|\Pi_{\text{Heeg}}| = |H_{840}| = 6$ and the W-mirror is a bijection of the residue ring, the equality of sets follows. □

We may now record the disjointness of the two sets, which is the algebraic counterpart to the geometric statement that the W-mirror does not have fixed points within the unit quadratic residues at this modulus.

Proposition 9. *The Mordell residues and the Heegner residues are disjoint as subsets of the unit group of integers modulo eight hundred and forty.*

Proof. The intersection $H_{840} \cap \Pi_{\text{Heeg}}$ is empty if and only if -1 does not lie in the quadratic residue set of the unit group. The quadratic-reciprocity criterion states that -1 is a quadratic residue modulo a positive integer n if and only if every odd prime divisor of n is congruent to one modulo four. The integer eight hundred and forty has the odd prime divisor three, which is congruent to three modulo four, and therefore -1 is not a quadratic residue. Disjointness of the two sets follows immediately. □

3.3 Frobenius exponentiation acts as an involution preserving both sets

The third theorem records the symmetry that Frobenius exponentiation imposes on the residue structure.

Theorem 10 (BES–13)**.** *The Frobenius operator $\hat{\Phi}$, defined by exponentiation to the twenty-ninth power modulo eight hundred and forty, acts as an involution on the multiplicative unit group. It preserves both the Mordell residue set and the Heegner residue set as sets, and the orbit structure within each is identical: two fixed points and two orbits of size two.*

Proof. The operator is an involution because the integer twenty-nine has multiplicative order two in the unit group of integers modulo eight hundred and forty. This follows directly from the identity $29^2 = 841 = 840 + 1$, which states that twenty-nine squared is congruent to one modulo eight hundred and forty.

The preservation of the Mordell set is established by direct computation. Within H_{840}, the fixed points of $\hat{\Phi}$ are one and one hundred and sixty-nine; the orbits of size two consist of the pairs one hundred and twenty-one with three hundred and sixty-one, and two hundred and eighty-nine with five hundred and twenty-nine. These four orbits, with two singletons and two doubletons, exhaust the six elements.

The Heegner set exhibits the parallel structure. The fixed points are six hundred and seventy-one and eight hundred and thirty-nine; the orbits of size two are three hundred and eleven with five hundred and fifty-one, and four hundred and seventy-nine with seven hundred and nineteen. □

The disjointness established in Proposition 9 together with Theorem 10 yields a clean structural picture. The W-mirror exchanges the two residue sets element-wise, while the Frobenius involution permutes each set internally without ever crossing between them.

3.4 The Goldbach anchor at twice the conductor

The fourth theorem records a numerical anchor at the integer twice the conductor of one hundred and seven, which we call S.

Theorem 11 (BES–3)**.** *The number of unordered Goldbach pairs of primes summing to two hundred and fourteen, denoted $G(S)$, equals eight, which is two raised to the power N_c.*

Proof. Direct enumeration yields the eight unordered pairs of primes summing to two hundred and fourteen: three with two hundred and eleven; seventeen with one hundred and ninety-seven; twenty-three with one hundred and ninety-one; forty-one with one hundred and seventy-three; forty-seven with one hundred and sixty-seven; eighty-three with one hundred and thirty-one; one hundred and one with one hundred and thirteen; and the diagonal pair one hundred and seven with one hundred and seven. The cardinality is eight, equal to 2^3. □

The diagonal pair is fixed under the radial offset operator $\hat{M}_K$, in the sense that the sum of the operator values on the two primes equals zero. This connects the Goldbach pair count at the framework anchor to the antisymmetric eigenspace of the radial offset, although the broader anchor pattern at Lucas-tower integers does not extend cleanly, as is discussed in Section five.

4 The decomposition pipeline

We turn now to the practical question of constructing explicit Egyptian fraction triples for hard primes. The approach factorises the search into three sequential operators.

4.1 The three-operator factoring

Given a hard prime p, the goal is to find positive integers x, y and z with $x \leq y \leq z$ such that the sum of the reciprocals equals four divided by p. The operator $\hat{X}$ chooses the integer x, drawing on atlas-derived constraints. The operators $\hat{Y}$ and $\hat{Z}$ then derive the remaining components in closed form, by enumeration of the divisor pairs of a quadratic discriminant.

The operator $\hat{X}$ admits two implementations. The first, denoted $\hat{X}_{\text{mod } 8}$, restricts the search to integers x congruent to zero modulo eight within the admissible range from the ceiling of p divided by four to three quarters of p. This recipe exploits the structural fact that every Mordell residue is congruent to one modulo eight, so that residuals of the form $4x - p$ acquire a clean component in the modulo-eight Chinese remainder factor when x is divisible by eight.

The second implementation, denoted $\hat{X}_{\text{boundary}}$, is invoked as a fallback when the modulo-eight recipe does not succeed within the admissible range. It performs an exhaustive boundary search over a window of three hundred consecutive integers above the lower bound. Empirically, this fallback is required only for a small finite set of exceptional primes.

The operators $\hat{Y}$ and $\hat{Z}$, once x is chosen, are defined by an algebraic identity. Reducing the residual difference of four over p minus one over x to lowest terms, with numerator a and denominator b, the constraint that one over y plus one over z equals a divided by b admits the equivalent form that the product of $ay - b$ and $az - b$ equals b squared. The problem reduces to enumerating divisor pairs of b squared and recovering y and z when the divisor reconstruction yields integers. This step is closed-form, in the sense that no openness remains: once x is given, the determination of valid pairs is a finite exhaustive procedure on a known integer.

4.2 Empirical performance

The pipeline has been applied to every hard prime less than five times ten to the seventh. The total count is ninety-three thousand four hundred and fifty-seven primes, distributed approximately uniformly across the six Mordell residue classes, with each class containing between fifteen thousand five hundred and thirty-nine and fifteen thousand six hundred and sixty-eight representatives.

The modulo-eight recipe succeeds on ninety-three thousand four hundred and fifty-four of these primes, that is on all but three. The three exceptions are two thousand five hundred and twenty-one, seven thousand six hundred and eighty-one, and thirty-three thousand six hundred and one. For each of these exceptions, the boundary fallback succeeds within a window of at most five integers above the lower bound. Consequently, the combined pipeline achieves complete coverage of the tested range.

The exceptions exhibit a common structural pattern. The smallest valid x for each is at most five integers above the ceiling of p divided by four, and the parity of x modulo eight is non-zero in each case. For all primes outside the exception set, the smallest valid x lies sufficiently far above the lower bound that an integer congruent to zero modulo eight falls within the admissible window.

4.3 The atlas anchors verified by computation

A small number of structural identities at framework integers are confirmed by the empirical record. The Goldbach anchor at twice the conductor, recorded in Theorem 11, holds tightly. The Ramsey number $R(3, N_c)$ equals twice N_c, and $R(3, N_{st})$ equals N_c squared; both identities are exact at the small Ramsey values for which classical computation is available. The unique solution of the Catalan equation, fixed by Mihailescu in 2002, takes the form $N_c^2 - 2^{N_c} = 1$. The Brocard equation has its largest known solution at $7! = (71 - 1)(71 + 1)$, with seven equal to the Lucas integer L_4.

These identities are descriptive rather than causal, and we do not claim that the framework integers force the corresponding values. Their utility lies in providing a small set of clean reference points against which empirical computations may be checked.

5 Conjectures arising from the evidence base

The empirical record supports a number of conjectures concerning the behaviour of the pipeline outside the verified range and concerning related arithmetic problems. We list the principal ones, classified by the strength of their supporting evidence.

5.1 Conjectures of strong empirical support

Conjecture 12 (BES–1: finiteness of the exception set)**.** The set of hard primes for which the modulo-eight recipe does not yield an Egyptian fraction decomposition equals the three primes two thousand five hundred and twenty-one, seven thousand six hundred and eighty-one and thirty-three thousand six hundred and one.

The evidence comprises ninety-one thousand eighty-seven hard primes in the interval from one million to fifty million, with no new exception found.

Conjecture 13 (BES–2: Andrica saturation)**.** The supremum of the difference of square roots of consecutive primes is attained at the pair seven and eleven, equivalently at the Lucas integers L_4 and L_5, and never improved.

This supremum has been observed to remain constant at the value of the square root of eleven minus the square root of seven across all tested ranges up to one hundred thousand.

5.2 Conjectures of intermediate strength

The Hardy–Littlewood ratio for the twin prime counting function is conjectured to converge to one as the upper bound tends to infinity, with oscillation around the limit rather than monotone descent. The boundary gap distribution for the Erdős–Straus pipeline is conjectured to satisfy the property that the smallest valid integer x lies asymptotically at distance tending to infinity from the lower bound. The Brocard equation is conjectured to have only the three solutions known classically.

5.3 Conjectures of weaker support

The Cramér gap conjecture is suggested by limited data to satisfy a stronger constant than the unit conjectured classically. The Ramsey number at the integer twenty-nine, which lies beyond the range computed classically, is conjectured to take a value related to the framework integer eight hundred and forty-one. The Lehmer totient conjecture admits a quantitative refinement consistent with the standard literature. These conjectures rest on small numbers of data points and are recorded for completeness rather than as candidates for proof.

6 Lean 4 formalisation

The four central theorems of Section three have been formalised in the Lean 4 proof assistant, using the mathlib library at version four point thirteen. Each theorem reduces to a finite computation on the residue ring of integers modulo eight hundred and forty, and consequently the proofs use the `native_decide` tactic, which compiles the decision procedure to native code and evaluates it. The development is contained in five source files: a definitions module that introduces the framework constants and residue sets; a Mordell module containing Theorem 6 and its corollaries; a Heegner module containing Theorem 8 and the disjointness proposition; a Frobenius module containing Theorem 10; and a Goldbach module containing Theorem 11. The total source comprises approximately two hundred lines, and the build is reproducible by any user with elan and lake installed.

The Lean formalisation does not extend to the empirical results of Section four, which involve unbounded numerical searches and are not amenable to a finite decision procedure in the same form. The conjectures of Section five remain open and unformalised.

7 Scope and limitations

The contribution of this paper is threefold. First, the structural identification of the Mordell hard residues with the quadratic residues of the unit group modulo eight hundred and forty, expressed in Theorem 6, replaces a list of six values with a clean algebraic statement that explains the count and the modular structure simultaneously. Second, the W-mirror connection of Theorem 8 establishes a bridge between the Erdős–Straus hardness and the Heegner geometry that arises in the elliptic-curve setting at the same conductor. Third, the three-operator decomposition pipeline of Section four provides a complete and verifiable algorithm for the construction of explicit Egyptian fraction triples on every hard prime in the tested range.

We do not claim a proof of the Erdős–Straus conjecture. The conjecture concerns every integer not less than two, whereas the empirical record extends to five times ten to the seventh and rests on numerical search. We do not claim that the pipeline extends to all hard primes; the strong empirical pattern documented here may break at scales beyond the tested range. The atlas anchors of subsection 4.3 are descriptive, not causal, and we make no claim that the framework integers determine the Ramsey or Catalan or Brocard values cited there. Finally, the conjectures of Section five are recorded with explicit strength labels, and we encourage readers to falsify them by extending computation or by analytical argument.

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Data and code availability

The empirical record described in Section four, along with the Lean 4 formalisation summarised in Section six, is available in the deposit accompanying this paper.

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